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THEORETICAL ADVANCES

A novel model for fall detection and action recognition combined lightweight 3D-CNN and convolutional LSTM networks

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Abstract
Three-dimensional convolutional neural networks (3D-CNNs) and full connection long short-term memory networks (FC-LSTMs) have been demonstrated as a kind of powerful non-intrusive approaches in fall detection. However, the feature extration of 3D-CNN-based requires a large-scale dataset. Meanwhile, the deployment of FC-LSTM to expand the input into one-dimension leads to the loss of spatial information. To this end, a novel model combined lightweight 3D-CNN and convolutional long short-term memory (ConvLSTM) networks is proposed in this paper. In this model, a lightweight 3D convolutional neural network with five layers is presented to avoid the phenomenon of over-fitting. To further explore the discrimnative features, the channel- and spatial-wise attention modules are adopted in each layer to improve the detection performance. In addition, the ConvLSTM is presented to extract the long-term spatial–temporal features of 3D tensors. Finally, we verify our model through extensive experiments by comprehensive comparisons with HMDB5, UCF11, URFD, and MCFD. Experimental results on the public benchmarks demonstrate that our method outperforms current state-of-the-art single-stream networks with 62.55 ± 7.99% on HMDB5, 97.28 ± 0.36% on UCF11, 98.06 ± 0.32% on URFD, and 94.84 ± 4.64% on MCFD.

Keywords Fall detection · Action recognition · Three-dimensional CNN · Attention module · Single-stream network

1 Introduction

The proportion of aging people keeps a steady rise recently, which has brought great challenges to the development of the world's public health system. As the largest developing country in the world, China is also difficult to escape. Actually, the proportion of elderly people is expected to rise up to 35% by 2050 [1]. In general, fall occurring indoors often results in serious consequences for the elderly owing to the poor indoor lighting, clutter, slippery floor, or unstable furniture [2–4], etc. What's more, the elderly living alone is vulnerable to be life-threatening since fall usually leads to concussion, hemorrhage, or other serious health risks. Therefore, it is a necessary to develop intelligent fall detection and monitoring systems, so as to detect falls effectively, inform the families promptly, and deliver medical service instantly for the old.

At present, extensive attention has been paid to fall detection from academia and industry [5, 6]. In general, the approaches for fall detection can be divided into two categories, *i.e.*, methods based on non-visual devices and those based on vision devices.

The non-visual devices consist of ambient sensors or wearable sensors. The ambient sensor-based approach utilizes sensing devices to gather sound, vibration, or other signals for fall detection, such as heart beat sensor, body temperature sensor, room temperature sensor, CO sensor, and CO_2 sensor, to monitor the basic vital signs in real time [7]. It mainly triggers the alarm setting by means of sensing the pressure generated by the subject around the environment. However, the pressure is not always caused by the fall of subjects, which leads to some uncertainty to fall detection [8]. The wearable sensor-based approach mainly uses various sensors to detect changes in gaits of human movements to evaluate the state of human body. A

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variety of studies focus on the collection of data via deep learning through the above methods, such as features from complicated accelerometer data automatically and simplifying data for pre-processing in a convolutional neural network (CNN) [9]. Different combinations of sensors are utilized to accumulate various data in a novel classification approach based on support vector machines (SVM), random forest (RF), and K-nearest neighbors (KNN) [10], such as wearable sensors, electroencephalograph helmet, infrared sensors, etc. Moreover, wearable accelerometers are utilized to gather data in real time for different recurrent neural networks (RNNs) [11]. Constraint wearable sensors are employed to collect data for lightweight and supervised CNN in some extreme scenarios [12]. As for the wearable sensor-based method, it is able to achieve an excellent accuracy. However, it may also result in uncertainty for fall detection if the sensor is unworn.

The method based on vision devices usually detects fall through extensive analysis on videos or images [13–18]. Specifically, fall detection algorithms based on deep learning become increasingly popular, such as a deep learning approach with skeleton-based 3D-CNN, and consecutive-low-pooling is utilized to obtain the discriminant skeleton representation for fall detection [19]. In addition, a great deal of studies concentrate on skeleton data gathered in deep learning model, such as skeleton data accumulated through OpenPose [20]. Furthermore, a new dataset of a human skeleton map is utilized to train a new model [21]. Moreover, the human skeleton estimation is utilized to obtain features extraction in a vision-based approach for fall detection and activity recognition [22]. Besides, the up-fall dataset is utilized as one of the test datasets [23].

Compared with the wearable sensor-based method, the method based on vision devices largely eases discomfort caused by invasive manner. Additional vision equipment is a necessary, but the price of these equipments will gradually decline with the development of technologies. In addition, deep learning algorithms are more effective than traditional handcraft detection approaches. However, the scale of 3D-CNN is larger, and the input fall dataset may have limited images and videos. Moreover, LSTM can only accept 2D tensor. Therefore, the reduction in the scale of 3D-CNN and using convolutional LSTM (ConvLSTM) as a substitute for LSTM is a necessary.

To this end, a novel model for fall detection and action recognition is proposed in this paper, which consists of three crucial modules, namely, a lightweight 3D-CNN with five 3D convolutional layers, a channel- and spatial-wise attention modules for 3D-CNN, and a convolutional LSTM. To be specific, the main contributions of this paper are presented as follows.

(1) A novel network is proposed for fall detection and action recognition. A combination of a lightweight 3D-CNN and ConvLSTM is presented to extract the features from video sequences. The representation of the spatial information is strengthened, and the fine-grained extraction of fall and action features is enabled.
(2) The channel- and the spatial-wise attention modules are elaborately designed to infer attention in turn along the channel and the spatial dimensions in shallow 3D-CNN. As a result, the fine-grained features can be effectively captured to obtain discriminative features by effective lightweight 3D-CNN.
(3) Our network is evaluated by extensive comparisons with multiple benchmarks UCF11, HMDB51, MCFD, and URFD. Experimental results show that our network consistently outperforms several state-of-the-art single-stream networks. Specifically, the proposed method achieves 100% sensitivity and 99.76% specificity on the URFD and MCFD databases, respectively.

The remainder of this paper is organized as follows. In Sec. II, a briefly introduction to CNN is presented, along with the attention mechanism and the ConvLSTM applied in fall detection. The detailed design of our proposed is presented in Sec. III. In Sec. IV, extensive experiments are conducted in terms of UCF11, HMDB51, MCFD, and URFD, and the discussion on experimental results is made. Finally, the conclusion is drawn, and the future work is pointed out in Sec. V.

2 Related works

2.1 Convolution neural network

Convolutional neural network (CNN) originated from the neocognitron model was proposed by Fukushima [24, 25]. Haut et al., believed that the operation mechanism of CNN is similar to the behavior of neurons in the primary visual cortex of biological brain [26]. The size of the area depends on that of the convolution kernel.

The theory of deep learning (DL) was put forward in 2006, and then, CNN has been obtained widely concerns [27]. A variety of approaches were proposed to improve the performance of 2D-CNN. However, as a kind of fall detection method based on videos, it only focuses on the motion feature between adjacent frames rather than a single frame. Therefore, it is inefficient to conduct fall detection or action recognition only with a single static frame of single-stream 2D-CNN. Therefore, Marcos et al., proposed a two-stream 2D-CNN combined RGB streams with the optical flow stream to incorporate the motion information with the optical flow [28]. However, the optical flow is time-consuming, and two streams of 2D-CNN require a higher GPU capacity owing to the large-scale parameters. As a consequence, 3D-CNN is adopted to extract features from both of the spatial and the temporal dimensions, and the 3D convolutional layers are utilized to capture the motion information encoded in multiple adjacent frames.

The conventional convolution operation of 3D-CNN is conducted spatially and temporally respectively. Guan et al., proposed a 3D-CNN and long short-term memory (LSTM) method with the adoption of the optical flow images and the motion history image [29]. The latest research indicates that the increasing depth of the network has both positive and negative effects. Feichtenhofer et al., introduced a slow-fast network which is considered as a combination of the fusion technique and 3D-CNN [30]. There are two streams, namely, a fast path which focuses on the spatial information as RGB stream and a slow path which regards the temporal information as an optical flow stream. Tran et al., proposed a channel-separated convolutional networks (CSN) which is constructed by the channel interactions and the spatial–temporal interaction, which is called as 3D-CNN version of depth-wise separable convolution [31]. Sudhakaran et al., proposed a few gating methods to upgrade the spatial–temporal decomposition of 3D convolution kernel and achieved the SOTA performance [32]. Xin et al., proposed a skeleton-based 3D consecutive-low-pooling neural network to process skeleton representation sequences [33].

The methods mentioned above mainly employed 3D-CNNs to explore the discritimative features by means of encoding motion information in multiple adjacent frames so as to further improve the accuracy. However, they failed to prevent trainable model from over-fitting caused by the large-scale network. In fact, most of researches based on deep learning leverage large-scale benchmarks to pre-train a network, and then, the network is fine-tuned during the process of training.

To this end, a lightweight 3D-CNN is proposed to circumvent over-fitting. Five the convolutional layers followed by a batch of normalization-RELUs (BN-RELUs) are presented as a lightweight 3D-CNN for feature extraction. As a result, the original frame is reshaped to the fixed 112×112.

2.2 Attention mechanisms

Attention mechanism plays an important role in computer vision, especially in object recognition and pose estimation. Recent years have witnessed that a great effort has been devoted to the network with attention mechanisms. Wang et al., proposed a residual attention network based on an attention module which utilizes an encoder–decoder structure [34]. The network is proven to perform well and be robust to noisy data through refining feature map. Hu et al., proposed a plug and play module named squeeze and an extraction block to improve the accuracy by means of modeling the correlation between the feature channel and the strengthening important feature [35]. This lightweight model can be widely utilized in various residual models with an excellent effect. Park et al., proposed a bottleneck attention module (BAM), which is mainly designed along the channel and the space dimension and can be embedded in various neural networks [36]. Woo et al., proposed a convolutional block attention module (CBAM) inferring attention in turn along with the channel and the spatial dimension [37]. This model also belongs to a lightweight plug and play model with a low cost and can be seamlessly integrated into the CNN architecture [38].

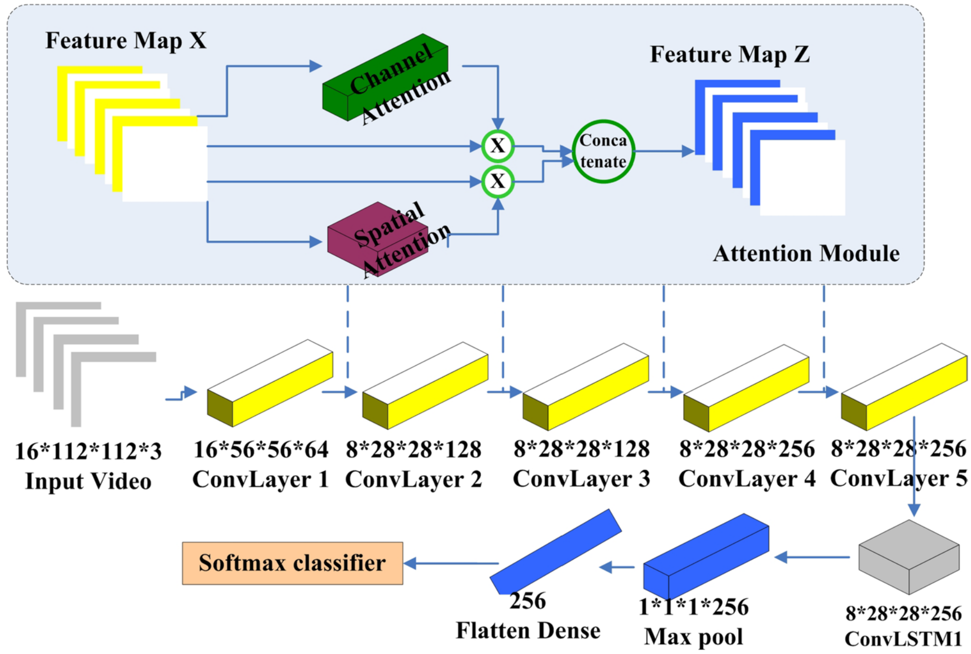

Fig. 1 The overall diagram of our proposed network. (Top) Applying channel-wise and spatial-wise attention modules in parallel to 3D-CNN for refining features. (Bottom) Utilizing lightweight 3D-CNN with ConvLSTM to process redundant spatial data

Table 1 The proposed architecture

Architecture	Channel	Kernel size	Ratio
Conv3D_1	64	$3\times7\times7$	–
Attention_1	64	$3\times7\times7$	8
Conv3D_2, 3	128	$3\times3\times3$	–
Attention_2, 3	128	$3\times3\times3$	8
Conv3D_4, 5	256	$3\times3\times3$	–
Attention_4, 5	256	$3\times3\times3$	8
ConvLSTM	256	3×3	–
Max-pooling	256	$8\times28\times28$	–

The methods mentioned above mainly employ the plug and play module to optimize the feature map and improve the classification performance. However, they are mainly designed for 2D-CNNs and fail to take the 3D-CNNs into consideration.

In this paper, we proposed a method based on three-dimensional convolution neural network (3D-CNN), the channel- and the spatial-wise attention modules, and convolutional long-term and short-term memory (ConvLSTM) network. A 3D-CNN is applied to encode both spatial information over each frame and the temporal information among different frames. Utilizing lightweight 3D-CNN and pretraining this model on Sports-1 M will be conducive to address the issue of over-fitting. Rather than the 2D-CNN, the channel- and spatial-wise attention modules are taken into consideration to improve the performance of 3D-CNN.

3 Methodology

In this section, the channel- and spatial-wise attention modules embedded in 3D-CNN are detailed. Subsequently, the architecture, a combination of a novel 3D-CNN with convolutional LSTM, is proposed for action recognition and fall detection. Finally, the loss function during for training is illustrated.

3.1 Lightweight 3D-CNN network architecture

The network architecture is depicted in Fig. 1. The detailed layer information is illustrated in Table 1.

The architecture has five convolution, five pairs of channel- and spatial-wise attention modules, one ConvLSTM, and one max-pooling layer, followed by a softmax classifier. The pool size of the max-pooling layer is the same as the stride to make the output size as $1\times1\times1\times256$.

Firstly, a three-dimensional cube which is composed of 16 frames with a size of 112×112 is fed into a lightweight 3D-CNN, and then, the channel- and spatial-wise attention modules are applied to the lightweight architecture to improve the effectiveness along with the reduction in the parameter overhead. The number of channels of the

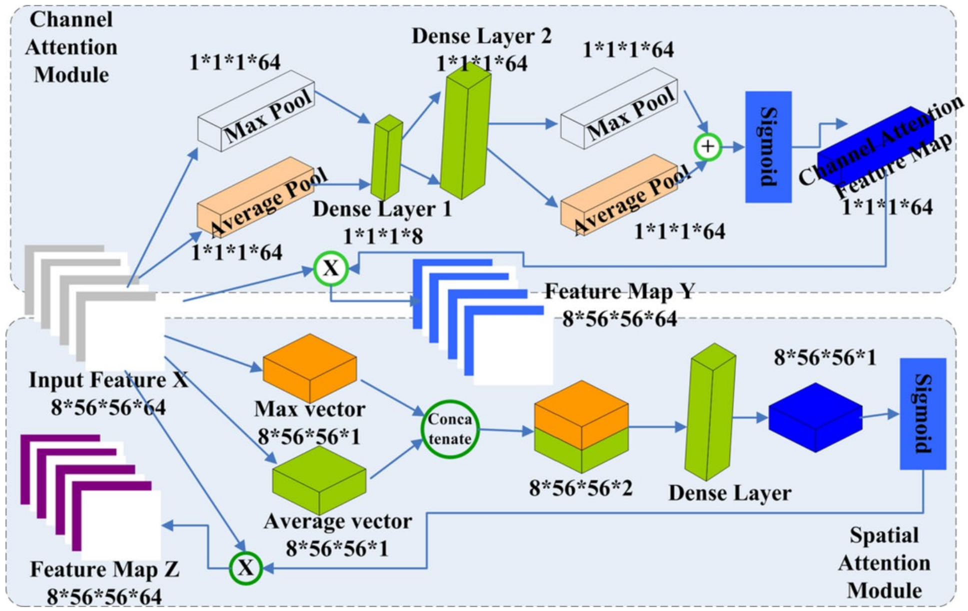

Fig. 2 Diagram of channel-wise and spatial-wise attention modules. (Top) Channel-wise attention module. (Bottom) Spatial-wise attention module. As illustrated, the channel-wise module uses both average- and max-pooling features with shared network composed of two dense layers followed by RELU. The spatial-wise module utilizes two similar features fed into a convolutional layer

Table 2 The architecture of channel- and spatial-wise modules

Architecture	Channel	Output shape	Ratio
Max-pooling_1,2	64	1 × 1 × 1 × 64	–
Average-pooling_1,2	64	1 × 1 × 1 × 64	–
Dense layer 1	8	1 × 1 × 1 × 8	–
Dense layer 2	64	1 × 1 × 1 × 64	8
Max-pooling_3	1	8 × 56 × 56 × 1	–
Average-pooling_3	1	8 × 56 × 56 × 1	–
Dense layer 4	1	8 × 56 × 56 × 1	–

five convolutional layers is set to be 64, 128, 128, 256, and 256, respectively, each of which is followed by batch normalization RELU (BN-RELU) function. Note that the first layer differs from the others in the size of kernel and stride, where the various convolution kernel sizes are $3 \times 7 \times 7$ and $3 \times 3 \times 3$, respectively, and the stride is fixed to (1, 2, 2) and (1, 1, 1), respectively.

Secondly, a pair of attention modules (described in Sec. III (A)) are appended between two convolutional layers. The channel-wise attention feature map is generated from a channel-wise attention module, and its output size is $1 \times 1 \times 64$. An average-pooling layer and a max-pooling layer are deployed before and after the first dense layer and the second layer, respectively. The output size of the average-pooling layer and that of the max-pooling layer is set to $1 \times 1 \times 64$. The output sizes of the two dense layers are set to $1 \times 1 \times 8$ and $1 \times 1 \times 64$, respectively. In the spatial-wise attention module, the average-pooling layer and the max-pooling layer are utilized before a dense layer. The output size of the average-pooling layer and the max-pooling layer is set to $56 \times 56 \times 1$. The output size of the dense layer is $56 \times 56 \times 1$. The structures of channel- and the spatial-wise attention modules are presented as shown in Fig. 2. The detailed layer information is illustrated in Table 2.

The architecture of channel-wise module has an average-pooling layer and a max-pooling layer followed by two dense layers. The architecture of the spatial-wise module has an average-pooling layer and a max-pooling layer followed by one dense layers.

Thirdly, instead of FC-LSTM, the ConvLSTM is utilized to overcome the missing of the spatial data resulted from unfolding the inputs to 1D vectors. All the related data of ConvLSTM are 3D tensors. Therefore, the intermediate feature tensors are put into the max-pooling layer; finally, the softmax function is adopted as the classifier.

The detailed training process of the proposed model is presented as shown in Algorithm 1.

Algorithm 1 The training process of the proposed model.

```
Input: Training Images X_train and annotationing Y_train with C
       classes, MaxEpoch, img_nums_train, batch_size
Output: Trained model with target branch and accuracy
/*Training */
Initialize model with weights of pre-trained model
while e<MaxEpoch do
  for i=0 to img_nums_train/batch_size+1 do
    From (X_train, Y_train), sample a batch setbatch;
    Compute loss;
    Update target branch;
    Compute accuracy;
    Compute average accuracy;
  e=e+1;
/*Testing */
Input: Testing images X_test and annotations Y_test with C classes,
       MaxEpoch, img_nums_test, weights of trained model
Output: testing accuracy
while e<MaxEpoch do
  for i=0 to img_nums_test/batch_size+1 do
    From (X_test, Y_test), sample a batch setbatch;
    Compute loss;
    Update loss;
    Compute accuracy;
    Compute average accuracy;
  e =e+1;
```

Specifically, the dropout technology is adopted, and the fixed dropout rate is set to 0.3 in FC layer to suppress the phenomenon of over-fitting.

The loss in the proposed model is defined as the sum of clip-level losses. Mathematically, the loss function is calculated as follows,

$$L(f(x), y) = L(\left(\sum_{i=1}^{S} f\left(x_i, y\right) = -\sum_{i=1}^{S} y \cdot \log f\left(x_i\right) \right) \tag{1}$$

where y denotes the true label and x_i the i-th the sequence image belonging to video x, $f(x_i)$ represents the learned features associated with sequence position i in a video sequence, and $f(x)$ the classification score calculated by classifier f. Finally, S is the total sequence of video.

For each video x annotated y, typical categorical cross entropy loss function is utilized to minimize the loss $L(f(x), y)$. In order to continuously update the parameters, the RMSprop method is adopted as an optimization function.

3.2 Channel- and spatial-wise attention modules

To improve the capability of feature extraction, the channel- and spatial-wise attention modules are applied in the lightweight 3D-CNN. The channel-wise attention module is adopted to focus on the intermediate feature map which is generated by the convolutional layer. Subsequently, a spatial-wise attention module is proposed to generate the spatial-wise attention map to locate the important region.

To be specific, the attention module consists of the channel- and the spatial-wise attention modules. Specifically, the input feature is put into both of the average-pooling layer and the max-pooling layers simultaneously. Then, the feature maps generated by the two pooling layers are put forward to the first dense layer and the second dense layer. As a result, two feature maps are obtained. Subsequently, the added feature map is established based on the two feature maps. The channel-wise attention feature map is generated after the sigmoid function. In addition, the feature maps generated by the average-pooling layer and the max-pooling layer in the spatial-wise module are concatenated as the concatenated feature map which is inputted into the dense layer. The detailed process is illustrated as follows.

With an intermediate feature map as the input, a channel-wise attention map is generated to focus on what is an input image. To compute the channel-wise attention effectively, the spatial information should be aggregated. The average-pooling layer is utilized in the attention module of squeeze and excitation (SE) block to compute the spatial feature map. The average- and the max-pooling features are adopted in CBAM to achieve a better inference of the channel-wise attention because the max-pooling layer can capture another important clue to improve the representation ability of the architecture. Based on our previous works, both two pooling layers are applied in the channel-wise attention module for 3D-CNN. The perfect performance of our proposal is experimentally demonstrated in Sec. IV.

As shown on the top of Fig. 2, the intermediate feature map X is squeezed from 3D-CNN to the average- and the max-pooling features by performing both pooling layers simultaneously firstly. Subsequently, the two different features are put forward into two dense layers as the shared network to produce the channel-wise attention map $M_c \in R^{c\times1\times1}$. The activation size is fixed to be c and $c/r \times 1 \times 1 \times 1$ $c/r \times 1 \times 1 \times 1$, respectively, at the first and second dense layers, where r and c refer to the reduction ratio and the number of channels. Subsequently, the two average- and max-descriptor vectors are merged through a element-wise summation operation. In a brief, the channel-wise attention map is calculated as follows,

$$M_c(X) = \sigma\left(\begin{array}{l} S(\text{Average Pooling}(X)) \\ +S(\text{Max Pooling}(X)) \end{array} \right) \tag{2}$$

where σ denotes the sigmoid function, and S denotes the dense layer.

A spatial-wise attention map is produced, which is the supplement for the channel-wise attention map, to focus on the significant region of an input image. To obtain the spatial-wise attention map M_S, the feature descriptor which is concatenated in parallel with the two descriptors generated by average- and max-pooling operations is applied to the convolutional layer. The specific computation process is illustrated as below.

Two 3D maps are generated, *i.e.*, $X_{\text{Average}} \in R^{1\times H\times W\times \text{Seq}}$ and $X_{\text{Max}} \in R^{1\times H\times W\times \text{Seq}}$, the average- and the max-pooling operations are utilized along with the channel axis. Then, the following convolutional operation is conducted to generate the 3D spatial-wise attention map M_S, which is multiplied by intermediate feature map X as the final outputs Z of the spatial-wise attention module. Specifically, the spatial-wise attention map is calculated as follows,

$$Z = \mathrm{X} \otimes M_s \tag{3}$$

$$M_S(X) = \sigma(f^{7\times7\times7}([\text{Average}(X);\text{Max}(X)])) \tag{4}$$

where $\otimes$ denotes element-wise product, σ denotes the sigmoid function, $f^{7\times7\times7}$ denotes the convolutional operation with the $7 \times 7 \times 7$ kernel size. *Average*(•) and *Max*(•) mean the average-pooling layer and max-pooling layer, respectively.

The input of the spatial-wise module is set to be the product of the intermediate feature map X and the 3D channel-wise attention map M_C, which is expressed as follows,

$$Y = X \otimes M_C \tag{5}$$

The arrangement of both attention modules is illustrated in the top of Fig. 1. The experimental results are discussed in Sec. IV(B).

4 Experiments and results analysis

4.1 Dataset and parameter settings

To evaluate the performance and applicability of the proposed network, our proposal is applied on publicly available UCF11 and HMDB51 for action recognition, multiple cameras fall dataset (MCFD) and UR fall detection dataset (URFD) for fall detection [39–42].

MCFD is recorded under 24 scenarios with eight video cameras, involving 192 videos and different actions, such as walking, falling, lying, sitting, standing, and so on. The size of each frame is set to 720×480. URFD contains 70 sequences, including 30 falls recorded by two cameras deployed on the ceiling and floor, respectively, and one accelerometer, 40 ADLs recorded by one camera and one accelerometer.

UCF11 from YouTube consists of 11 kinds of actions (biking, golf, riding, and walking along with dog, etc.), including 1600 clips divided into 25 groups and similar characteristics in each group. HMDB51 is large human motion database, including 6849 samples, covering five classes and 51 sub-classes. Each sub-class is comprised of 101 videos.

In the experiment, the publicly available datasets are all divided into the training and the testing sets with 70 and 30%, respectively, for previously four datasets, 16 frames are selected according to the sampling interval. Then, each frame image is reshaped into a fixed size of 112×112. Rmpsprop optimization function is adopted during all the experiments, and the batch size and epoch are fixed to 32 and 60, respectively. The learning rate is initially set to 0.1, and it will decrease when the accuracy of the model fails to be improved after the 25th epoch.

To thoroughly evaluate the effectiveness of the proposed network, extensive ablation experiments are designed for the channel- and spatial-wise attention modules. To further demonstrate the performance, extensive comparisons are made between our network and other state-of-the-art approaches with regard to MCFD and URFD. Note that 3D-CNN is pretrained on Sports-1 M to circumvent over-fitting.

4.2 Ablation experiments for channel- and spatial-wise attention modules of 3D-CNN

In this section, the effectiveness of our model and the classification accuracy on the validation set are empirically reported. For the ablation study, UCF11 is used, and five 3D convolutional layers, 256-dimension ConvLSTM, and the max-pooling layer are adopted as the baseline. Three experiments are designed to explore the effective of the proposal, which are detailed as follows.

Firstly, we experimentally demonstrate the adoption of the average-pooling layer and the max-pooling layer in both attention modules. Then, we compare three different utilization approaches of average-pooling layer and max-pooling layer of channel-wise module, namely, only the average-pooling layer, only the max-pooling layer, and combination of the both two pool layers. With the adoption of both pooling layers, two dense layers are introduced to reduce the number of parameters. In this experiment, the reduction ratio is set to be 8.

Using average-pooling and max-pooling features performs best.

Table 3 presents the experimental results with various constructions of channel attention. Note that max- and average-pooling features can improve the accuracy compared with the baseline. In addition, it indicates that the channel-wise attention with average-pooling layer performs slightly better than that of the module with max-pooling layer, and using both max-pooling layer and average-pooling layer will perform best. In the following experiments, we apply both features simultaneously in 3D-CNN and merge the outputs of two dense layers through element-wise summation. The bold in Tables 3 to 11 denotes the accuracy of this architecture is best compared with that of other architectures in the table.

Given the channel-wise refined features, an effective approach is proposed to construct the architecture of the spatial-wise attention which is symmetric with the channel-wise attention branch. To generate a 3D spatial-wise attention feature map, a concatenated 3D descriptor that encodes the channel-wise refined features information is generated. Subsequently, one convolutional layer is applied to the 3D descriptor, obtaining the final attention map which is normalized by the sigmoid. Finally, the spatial-wise attention map is obtained.

Secondly, the effect of a kernel size at the convolution layer is designed. To be specific, the connected 3D descriptor is generated by means of utilizing the average- and the max-pooling layers across the channel axis. In addition, we study the effect of a kernel size at the convolution layer, namely, kernel size of $3\times3\times3$ and $7\times7\times7$. In the experiment, the spatial-wise attention module is placed after the channel-wise attention module, as both of the modules will be used simultaneously in the following experiments.

Using channel pooling method and the convolutional layer along with kernel size of 7 were to produce the best accuracy.

The experimental results are shown in Table 4. Note that using the average- and the max-pooling layers generate a better accuracy, implying that poolings lead to a finer attention inference. It can be observed that adopting a larger kernel size produces a better accuracy in the case of employing both of max- and average-pooling layers and channel-wise attention. It means that a larger receptive field plays an important role in spatially important regions. As a result, we adopt the average- and the max-pooling layers along the channel axis with kernel size of $7\times7\times7$ as our spatial attention module.

Thirdly, a combination of the channel- and the spatial-wise attention modules is utilized to explore an effective approach. In the experiment, there are three different ways to arrange both modules, sequential channel–spatial, sequential spatial–channel, and channel–spatial in parallel. They are compared with regard to the accuracy. It is well known that each module has a different function. As a result, the arrangement of modules may affect the overall performance of the whole model. Consequently, the two modules can combine in parallel, namely, connecting the outputs of the two attention modules.

Using both the channel- and the spatial-wise attention modules are important and the best-arranging strategy (*i.e.*, in parallel) further pushes performance.

Based on the experimental results on different arranging ways summarized in Table 5, it can be concluded that both attention modules arranging in parallel outperform arranging in sequential order, demonstrating that the increasing parameters in the network has positive effect. It implies that

Table 3 Comparison of different channel-wise attention methods based on UCF11

	Architecture	Accuracy(%)
1	Base_model	93.02 ± 5.25
2	Base_model + channel(AvgPool)	95.9 ± 0.9
3	Base_model + channel(Maxpool)	95.62 ± 0.94
4	Base_model + channel(AvgPool&MaxPool)	$\mathbf{96.06\pm0.46}$

Table 5 Comparison of different combining ways of channel and spatial attention based on UCF11

	Architecture	Param	Accuracy(%)
1	Base_model + channel + spatial	3.235 M	96.56 ± 0.56
2	Base_model + spatial + channel	3.235 M	95.76 ± 0.46
3	Base_model + channel&spatial in parallel	**6.470 M**	$\mathbf{97.28\pm0.36}$

Table 4 Comparison of different spatial-wise attention methods based on UCF11

	Architecture	Accuracy(%)
1	Base_model	93.02 ± 5.25
2	Base_model + channel	96.06 ± 0.46
3	Base_model + channel + spatial($1\times1\times1$ conv, kernel_size = $3\times3\times3$)	96.11 ± 0.3
4	Base_model + channel + spatial($1\times1\times1$ conv, kernel_size = $7\times7\times7$)	96.13 ± 0.26
5	Base_model + channel + spatial(avg&max, kernel_size = $3\times3\times3$)	96.18 ± 0.11
6	Base_model + channel + spatial(avg&max, kernel_size = $7\times7\times7$)	96.56 ± 0.56

generating an attention map in parallel infers a finer attention map than doing sequentially. In addition, the channel first order performs slightly better than the spatial first order in 3D-CNN. All these show that the best-combining approach for both attention modules further improve the accuracy.

Throughout the ablation experiment, we explore the effectiveness of the channel- and the spatial-wise attention modules in 3D-CNN. The max- and the average-pooling layers are chosen for both the channel- and the spatial-wise attention modules. A convolutional layer with kernel size of 7 is utilized in the spatial-wise attention modules. The arranging strategy which consists of both attention modules in parallel is introduced. Our final channel- and the spatial-wise attention modules combined with 3D-CNN is shown in Fig. 1. It achieves the best accuracy of 97.28 ± 0.36%, as shown in Table 5.

We experimentally verify that the parallel manner performs better than the sequential one. It can be found that the parallel arrangement achieved a better performance compared to the sequential arrangement at the cost of larger parameter overhead in 3D network in our experiment.

4.3 Experiments on UCF11 and HMDB51

Classification experiments on UCF11 and HMDB51 are conducted to rigorously evaluate our model. The channel- and spatial-wise attention modules are applied in 3D-CNN. The experiments follow the specification illustrated in Sec. IV(B), and the performance of the various architectures of our model is evaluated objectively.

In the experiment, ConvLSTM is deployed after lightweight 3D-CNN. The effect of the different number of dimensions is explored in ConvLSTM. Instead of the full connection, the average-pooling or the max-pooling layer is applied after the ConvLSTM. The softmax is adopted as the classifier. Different sampling intervals are set to search for the best-sampling strategy. To obtain a preciser performance, a fivefold cross-validation scheme is proposed. Specifically, the dataset is divided into five subsets with the equal size, one of which is used for testing and other four for training.

The performance is evaluated in terms of the accuracy. The number of frame between two images sampled from video refers to the sampling interval in the following experiments. The effectiveness of max- and average-pooling layers is explored. In addition, we compare three different sampling intervals which are set 1, 2, and 4, respectively. The channel of ConvLSTM is set to 256. Moreover, we investigate the effect of different number of channels in the case of using the max-pooling layer, and the number of channels of ConvLSTM is set to 24, 64, 128, and 256, respectively.

Experimental results with various networks are shown in Table 6. Note that the model with average- and max-pooling layers achieves a better performance compared with the model with full connection layer. Note that max-pooling layer further outperforms the average-pooling layer, which indicates that explicitly max-pooling feature achieves a finer feature map than average-pooling feature. To the best of our knowledge, the more informative the input features is, the higher accuracy the network achieves. Therefore, we infer that sampling strategy may have a great impact on the performance. By comparison of different sampling schemes, we obtained that the model will perform the best when the sampling interval is set to 1. Specifically, the average accuracy has decreased by 1.38% with the increase in the sampling interval. It means that discriminative information of images may get lost owing to the increase in sampling interval. Therefore, we adopt the max-pooling layer. Moreover, according to the experimental results with different channels of ConvLSTM, the network with 256-dimension ConvLSTM significantly outperforms those with all the remaining dimensions, which means that the performance can be improved as the computational complexity increase without over-fitting. In a brief conclusion, the 256-dimension ConvLSTM is deployed after lightweight 3D-CNN, and the max-pooling layer is utilized.

Table 6 Comparisons on the performance with various network architectures based on UCF11

	Architecture	Accuracy(%)
1	(1 frame)3D-CNN + ConvLSTM(256)	94.3 ± 0.6
2	(1frame)3D-CNN + ConvLSTM(256) + Averagepool	95.6 ± 0.85
3	(1 frame)3D-CNN + ConvLSTM(256) + Maxpool	**97.28 ± 0.36**
4	(2 frame)3D-CNN + ConvLSTM(256) + Maxpool	95.9 ± 1.07
5	(4 f rame)3D-CNN + ConvLSTM(256) + Maxpool	94.98 ± 1.41
6	(1 frame)3D-CNN + ConvLSTM(24) + Maxpool	95.03 ± 0.99
7	(1 frame)3D-CNN + ConvLSTM(64) + Maxpool	95.05 ± 0.36
8	(1 frame)3D-CNN + ConvLSTM(128) + Maxpool	95.27 ± 1.89

Using the proposed 3D-CNN (with channel- and spatial-wise attention modules) along with ConvLSTM, one frame, two frames, and four frames all mean the number of frame between two images sampled from videos.

The model consisting of three parts is designed, following the protocol of the channel- and spatial-wise attention modules as specified in Sec. IV(B) to construct 3D-CNN. The 256-dimension ConvLSTM and max-pooling layer are applied. Our final model is presented as shown in Fig. 2. It achieves the best accuracy of 97.28 ± 0.36%, as shown in Table 5.

For better illustration, Table 7 summarizes the comparison on the proposed method versus some state-of-the-art methods on UCF11. For a fair comparison, the ratio between the training set and the testing set is fixed to be the same in the experiments, as shown in Table 7. Moreover, the average

Table 7 Comparison of the proposed method versus some state-of-the-art methods based on UCF11

	Architecture	Param	GFLOPs	Accuracy(%)
1	Our model	6.470 M	0.0293	**97.28 ± 0.36**
2	VGGNet with LSTM soft attention [43]	–	–	84.87 ± 0.53
3	Visual attention guided 3D-CNN [44]	–	–	92.01 ± 0.35
4	3D-CNN with LSTM average pool [44]	–	–	84.27 ± 0.36
5	GoogleNet with average pool [45]	–	–	82.56 ± 0.98
6	GoogleNet with max pool [45]	–	–	81.60 ± 1.75
7	GoogleNet with LSTM soft attention [45]	–	–	84.96 ± 0.32
8	KCP-LSTM [46]	0.135 M	0.0154	87.5
9	PARNet [47]	–	64.9	97
10	CNN wih ensemble learning [48]	–	–	87.43

classification accuracy is adopted as the evaluation criterion. The model is trained on UCF11 in 14 s every epoch. It is observed that 3D-CNN with the channel- and spatial-wise attention modules combined with ConvLSTM and max-pooling layer outperforms the remaining architectures in Table 7, and our proposal improves the accuracy of the visual attention-guided 3D-CNN algorithm by 5.27%. Note that the networks based on 3D-CNN generally achieve the better accuracy than the methods based on 2D-CNN, which means that 3D-CNN with attention is benefit to obtain the finer feature map compared with 2D-CNN. The amount of parameters and the GFLOPs are both smaller than that of our proposal. However, the accuracy of this algorithm is lower than our proposal. In PARNet method, it is observed that the accuracy is lower than ours based on UCF11. It indicates that this method is not benefit to conducting fall detection based on the small-scale dataset. In the traditional CNN with ensemble learning approach, the amount of parameters are both lower than that of the proposed method. It is concluded that the traditional single deep learning network is not sufficient for improving the fall detection.

The experiments based on HMDB51 are performed to illustrate the widely applicability of model. In the experiment, the impact of different number of dimensions is evaluated in ConvLSTM of our network. The model is trained on HMDB51 in 84 s every epoch. During the experiment, the 3D-CNN along with the channel- and spatial-wise attention and max-pooling layer is adopted as the baseline. We further explore the effect of the different numbers of dimension in the following ConvLSTM, namely, dimensions of 128, 256, and 512. The classification accuracy is adopted as the metric for performance evaluation.

Using the ConvLSTM is crucial and the 256-dimension ConvLSTM further pushes the performance of the proposed model.

It can be obtained that the accuracy of our proposal is the best, indicating its superiority of our proposal to other models in Table 8. Compared with 2D-CNN, 3D-CNN is experimentally verified to extract finer feature map. The

Table 8 Comparisons on different dimensions of ConvLSTM based on HMDB51

	Architecture	HMDB51 (128)	HMDB51 (256)	HMDB51 (512)
1	Our model	**52.55 ± 8.84**	**62.55 ± 7.99**	**59.20 ± 3.25**
2	VGGNet with LSTM soft attention [43]	–	–	44.02 ± 0.59
3	Visual attention-guided 3D-CNN [44]	–	–	50.65 ± 0.39

method of visual attention-guided 3D-CNN improves the accuracy compared with the approach of VGGNet with LSTM, which means that visual attention module achieves a better effectiveness than LSTM. In addition, it can be seen that the attention mechanism achieves a better performance. As a result, the utilization of the channel- and spatial-wise attention modules in 3D-CNN is crucial.

As shown in Table 8, the performance of the proposed model is improved with the increase in the dimension of ConvLSTM, which means that the improvement of the accuracy is at the cost of a high computational complexity. The adoption of the 256-dimension ConvLSTM slightly outperforms the 512-dimension ConvLSTM by 3.35%, which means that the 512-dimension ConvLSTM usually leads to the phenomenon of over-fitting. Consequently, we adopt 3D-CNN with the channel- and spatial-wise attention modules, followed by the 256-dimension ConvLSTM.

The comparison between the proposed methods and the state-of-the-art algorithms is presented in Table 9. It can be observed that our network obtains a better performance than others. The accuracy of CoCLR integreted with CMMC is 4% lower than our lightweighted model. Moreover, the GFLOPs are higher than that of our method. However, the CoCLR with CMMC is utilized in fall detection and obtains the accuracy of 59% in the table. It can be concluded that

Table 9 Comparisons on the proposed method versus some state-of-the-art methods based on HMDB51

	Architecture	Param	GFLOPs	HMDB51
1	Our model	6.470 M	0.0293	**62.55 ± 7.99**
2	ActorSL [49]	–	–	46.1
3	CoCLR + CMMC [50]		22	59
4	VARD [51]	–	–	34.9

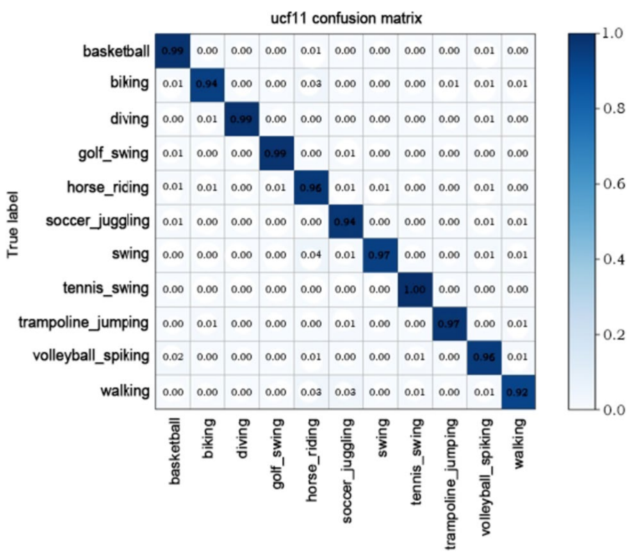

Fig. 3 Confusion matrix UCF11 of the proposed method

the good performance is benefit from the augmented samples generated by combining different modalities in videos enhance the quality of learned video representations. As shown in Table 9, the approach ActorSL effectively improves the performance and obtains the better accuracy than that of VARD. It is thought that is because the localized actors and their corresponding scene information are aligned and help the model to learn the descirminative regions and the background effectively. It can be obtained that VARD obtains the lowest accuracy.

Figures 3 and 4 depict the confusion matrix of the proposed network trained on UCF11 and HMDB51. It indicates that our network exhibits a high classification accuracy in UCF11 in Fig. 3. While for HMDB51, the performance is relatively poorer due to the large number of activity classes in Fig. 4.

4.4 Experiments on MCFD and URFD

Experiments on multiple camera fall dataset (MCFD) and UR fall detection dataset (URFD) are conducted to rigorously evaluate the generalization performance of the proposed model. For the two datasets, all the images are divided into three categories, namely, walking, falling, and lying. For the sake of convenience, they are labeled as 0, 1, and 2, respectively.

In these experiments, the proposed network is evaluated in fivefold cross-validation. A quantitative comparison with some state-of-the-art approaches is made in terms of the metrics of the sensitivity, the specificity, and the accuracy. The sensitivity shows the capability of correctly detecting the fall, and the specificity implies that of correctly detecting the non-fall. The sensitivity and specificity are calculated as follows:

$$\text{Sensitivity}: \ \mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{6}$$

$$\text{Specificity}: \ \mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{7}$$

$$\text{Accuracy}: \ \mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{8}$$

where TP (true positive) denotes the number of detecting fall accurately, and TN (true negative) means the number of accurately detecting non-fall. While FP (false positive) represents the number of detecting fall incorrectly, and FN (false negative) denotes the number of incorrectly detecting non-fall.

RGB, OF, and PE in parentheses denote RGB images, optical flow images, and pose estimate information, respectively. The values on the left and the right of symbol (±) are average and variance of sensitivity, specificity, and accuracy, respectively. The best result of single experiment is given in the parentheses.

The fall detection experiment is conducted on publicly available dataset URFD, following the same architecture as illustrated in HMDB51 experiment. The model is trained on URFD in 2 s every epoch. The experimental results based on fivefold cross-validation are summarized in Table 10. Hence, in addition to using average and variance, the accuracy is given in parentheses. It can be observed that RGBs are as effective as the optical flows with the input of single-stream, both sensitivities achieve 100%. However, the optical flows pre-processing is time-consuming, and only RGBs are hardly to improve the good accuracy and specificity of a single-stream model. It can be obtained that multi-stream model outperforms the single-stream approach in specificity and accuracy, which means that the integration of the optical flow (OF) and the pose estimate (PE) into RGB stream performs better. As a single-stream model, our architecture generates the best sensitivity 100% and specificity 99.15%, while the accuracy is slightly lower than those of the multi-stream model which gains more informative features by means

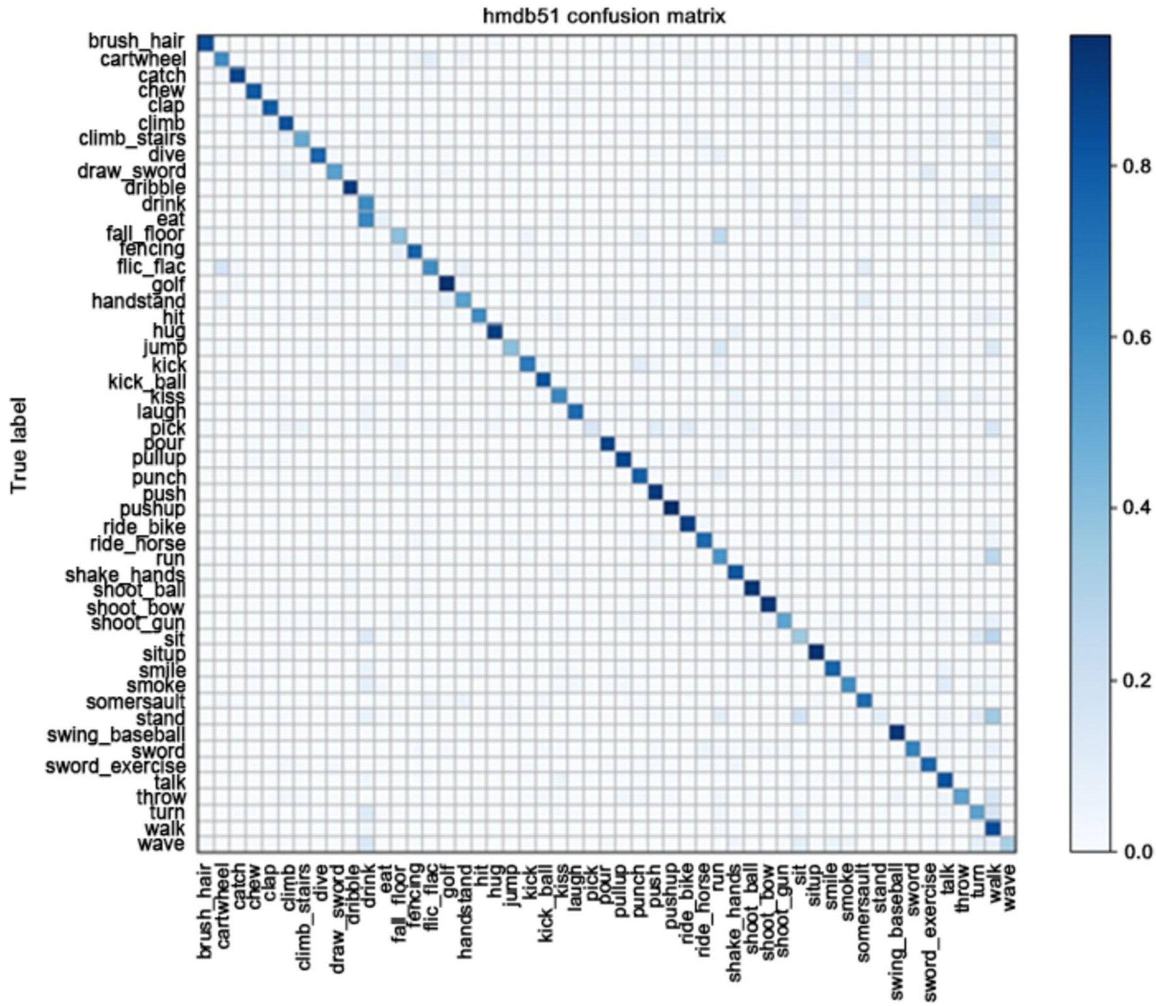

Fig. 4 Confusion matrix HMDB51 of the proposed method

Table 10 Comparisons on the proposed method with other state-of-the-art approaches based on URFD

	Architecture	TPR (sensitivity)	TNR (specificity)	Accuracy(%)
1	Our model	98.07 ± 0.3 **(100)**	**99.03 ± 0.15 (99.15)**	**98.06 ± 0.32 (98.3)**
2	Single stream (RGB) [52]	**100**	96.61	96.99
3	Single stream (OF) [52]	**100**	96.34	96.75
4	Multi-stream (RGB + OF + PE) [52]	**100**	98.61	98.77
5	EfficientNet-B0 [53]	93.33	100	97.14
6	Improved YOLOv5s [54]	–	–	97.2
7	A single frame human binary image with YOLOv5s [55]	–	–	96.7

of fusing various streams. Though the accuracy of the improved YOLOv5s is lower than ours, it is increased by 3.5% compared to YOLOv5s. The method based on a single frame human binary image achieves the lowest performance, which means that only innovating the method of processing image data will be not enough for boosting the performance. Finally, our architecture achieves the average sensitivity 98.07 ± 0.3%, the average specificity 99.03 ± 0.15%, and the average accuracy 98.06 ± 0.32%.

To further verify the performance of our network, the experiments on multiple cameras fall detection (MCFD) are designed with fivefold cross-validation, and the sampling interval is set to 1, 2, and 4, respectively. The model is trained on MCFD in 6 s every epoch. In addition, the

Table 11 Comparisons on the our proposal with other state-of-art approaches based on MCFD

	Architecture	TPR (sensitivity)	TNR (specificity)	Accuracy(%)
1	Our model (1 frame)	96.06 ± 1.9(97.96)	98.01 ± 0.88(98.89)	96.04 ± 1.64(97.68)
2	Our model (2 frames)	**96.88** ± 1.49(98.38)	**98.44** ± 0.68(99.12)	**96.92** ± 1.29(98.21)
3	Our model (4 frames)	94.91 ± 4.45(**99.36**)	97.40 ± 2.36(**99.76**)	94.84 ± 4.67(**99.51**)
4	Vision based (optical flow) [28]	99.0	96.0	–
5	Weakly supervised learning [56]	95.4	98.9	–
6	Grassmann manifold [57]	**100**	97	–

sensitivity, the specificity, and the accuracy are adopted. The results are listed as shown in Table 11.

The values on the left and the right of symbol (±) are average and variance of sensitivity, specificity, and accuracy, respectively. The best result of single experiment is given in the parentheses.

Our architecture is compared with the methods based on hand-crafted features and vision-based deep learning. It is obvious that the approach based on hand-crafted features achieves the best sensitivity 100%. The algorithm detects fall from a single camera and makes analysis on Grassmann manifolds. This approach outperforming our single-stream network in sensitivity. However, its specificity is lower than that of our proposed deep learning model. It can be obtained that our proposal achieves the better sensitivity and specificity compared with the vision-based approach. Hand-crafted low-level features can usually work well in some recognition scenarios. However, the high-level feature extracted by deep learning possesses general applicability across different recognition operations. To explore the appropriate sampling interval, we compute the average, the variance of the sensitivity, the specificity, and the accuracy, which are listed as shown in Table 11. Training with two sampling interval frames performs the best average value with the average sensitivity 96.88%, the average specificity 98.44%, and the average accuracy 96.92%, respectively. However, adopting four interval sampling frames brings the best single value with the sensitivity 99.36%, the specificity 99.76%, and the accuracy 99.51%, respectively. That is because our architecture with RGBs as input is more convenient and widely applicability in single-stream network. It is obtained that weakly supervised learning achieves the sensitivity 95.4% and specificity 98.9%, which are lower than other algorithms and our single-stream network, demonstrating that weakly supervised learning approach is ineffective for focusing on key region for fall detection.

4.5 Gradient-based visualization in 3D-CNN

For qualitative analysis, the visualization method Grad-CAM is applied to the proposed model which utilizes the images from UCF11, MCFD, and HMDB51 [58]. Note that network is able to focus on the important region during fall detection and action recognition. Figure 5 illustrates the visualization results of the images from UCF11, MCFD, and HMDB51 where the top rows are the original images, the middle rows are gray thermal diagrams, and the bottom rows are the color thermal diagrams. As we can see, the larger the weight, the darker the color of the relevant pixel.

In Fig. 5, note that attention regions of the proposed network in each image cover the target object regions in the bottom rows, implying the network learns well to exploit the informative target object region. Consequently, the appropriate attention mechanism is benefit to improving the classification performance eventually.

5 Conclusions

In this paper, we propose a novel network for effective fall detection and action recognition. It consists of two main modules, namely, a lightweight 3D-CNN and ConvLSTM. The former accurately characterizes the key information for each action feature by taking advantage of the channel- and the spatial-wise modules. In particular, the channel-wise module encodes the degree of the most salient part along the channel axis and obtains a channel-wise feature map. The spatial-wise module encodes the spatial-wise refined features information and generates a concatenated 3D descriptor. The latter effectively models the spatial correlation information and the long-term spatial–temporal features for each feature. Finally, we conduct extensive experiment to verify our proposal. The experimental results on UCF11, HMDB51, MCFD, and URFD databases exhibit the superiority of our proposal in performing falls detection and action recognition.

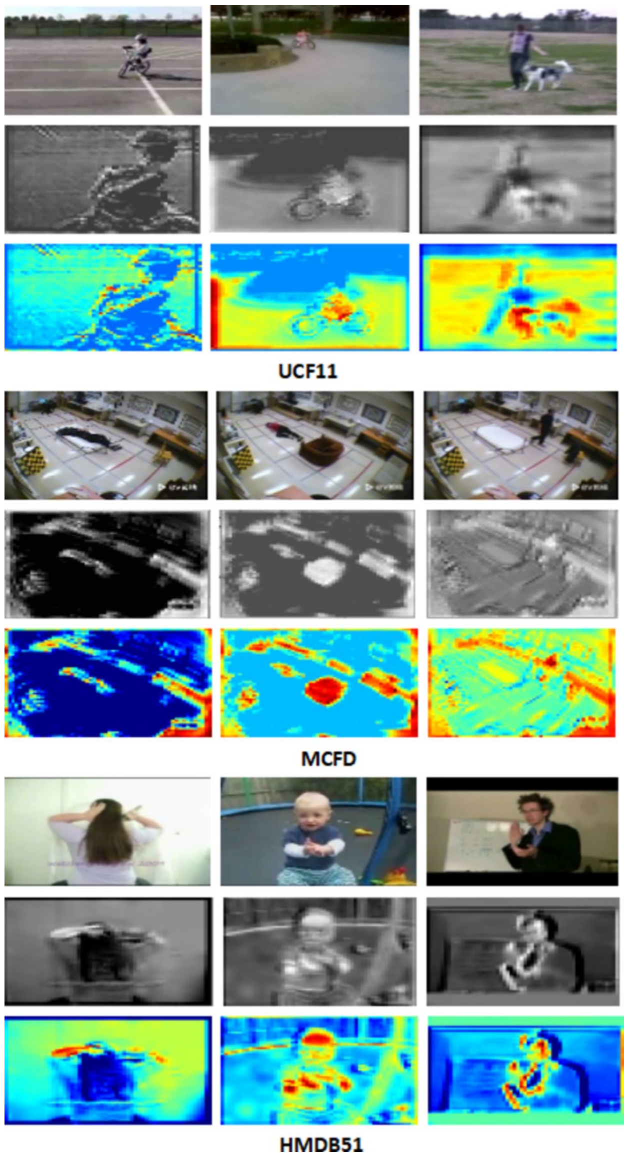

Fig. 5 Visualization results. In each sub-figure, the top rows are the original images, the middle rows are gray thermal diagrams, and the bottom rows are the color thermal diagram. Attention region located by the model combining 3D-CNN which channel- and spatial-wise attention modules are integrated into with ConvLSTM and max-pooling. The visualization is calculated for the last convolutional outputs

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Data availability Some or all data, models, or code generated or used during the study are available from the corresponding author by request.

Declarations

Conflict of interest All authors declare that there is no conflict of interest.

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